Case Study: Himachal Pradesh Heat Action Plan

High-Altitude Climate Adaptation, Ecosystem Resilience, and Mountain Disaster Governance

Policy Analysis Report

July 30, 2026

Abstract

This case study evaluates the Himachal Pradesh Heat Action Plan (HAP), analyzing how a high-altitude Himalayan state adapts heat resilience strategies to its topography, fragile ecology, and changing regional climate. Formulated by the Himachal Pradesh State Disaster Management Authority (HPSDMA) in coordination with the IMD and NDMA, the plan addresses heat risks across two distinct ecological zones: low-altitude valley belts experiencing extreme plains-like temperatures ($> 40°C$), and fragile high-altitude mountain ecosystems facing rapid thermal stress, springhead depletion, and cascading wildfire hazards.

1 Background & High-Altitude Climate Dynamics

Historically considered a cool refuge from northern India's summer heat, Himachal Pradesh has experienced significant pre-monsoon temperature spikes between April and June.

1.1 Emerging High-Altitude Hazards

- **Low-Elevation Heat Belts:** Districts in lower foothill belts—including Una, Hamirpur, Bilaspur, Kangra (Nurpur/Indora), and Solan (Nalagarh/Baddi)—regularly record summer temperatures exceeding 40°C to 44°C.
- **High-Altitude Thermal Anomaly:** Mid-to-high altitude regions (Shimla, Kullu, Solan) face accelerated warming trends, reduced winter snowpack, and diminished chill hours required for native horticulture.
- **Cascading Mountain Multi-Hazards:** Unlike flatland heatwaves, extreme heat in the Himalayas triggers secondary ecological hazards, including forest fires (*wildfires*), accelerated glacier/snowmelt, drying of mountain springs (*Naulis* and *Khatris*), and agricultural loss.

2 Institutional Framework & Core Pillars

The HPSDMA coordinates heat response actions across multiple specialized line departments:

2.1 Pillar 1: Terrain-Differentiated Early Warning Systems (EWS)

The India Meteorological Department (IMD) applies lower absolute temperature thresholds for declaring heatwaves in hilly regions compared to plains:

$$\text{Heatwave Criteria (Hilly Areas)} \iff T_{\max} \geq 30°\text{C} \quad \text{or} \quad \Delta T_{\text{anomaly}} \geq +4.5°\text{C} \tag{1}$$

Alerts are classified into **Yellow**, **Orange**, and **Red** risk levels and transmitted via the HPSDMA *Sachet* alert network, SMS pushes, and local radio broadcasts.

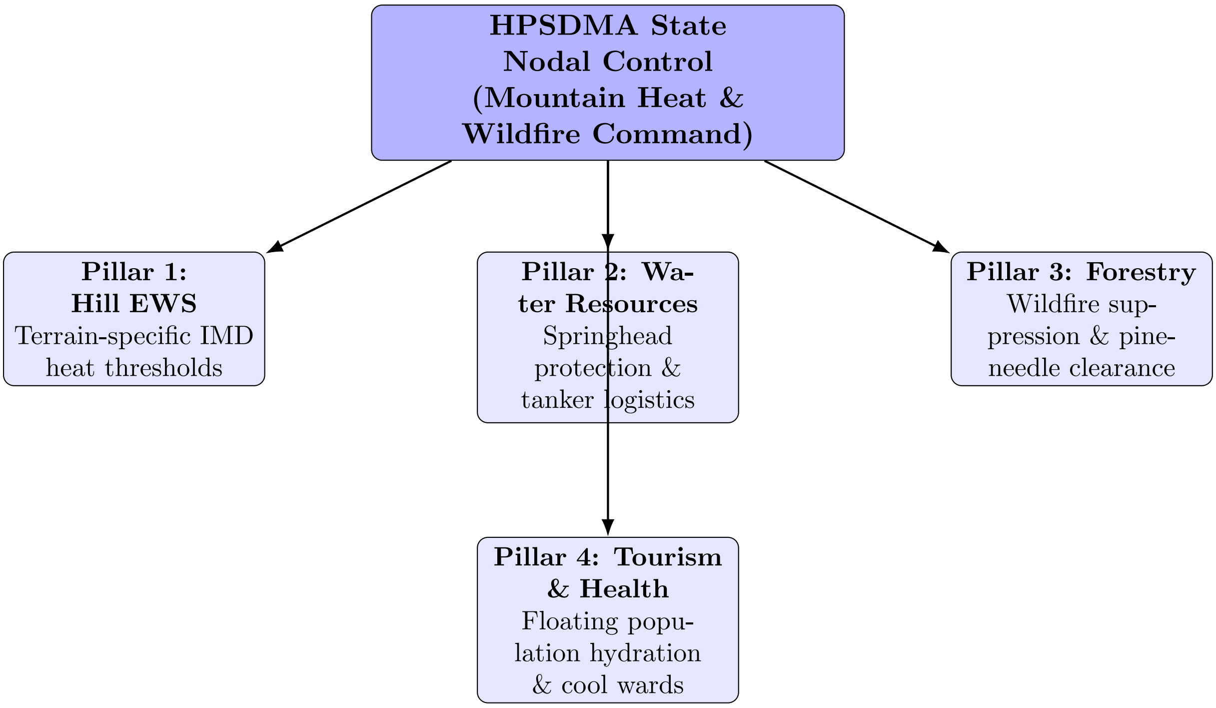

Figure 1: Systemic Architecture of the Himachal Pradesh Heat Action Plan.

2.2 Pillar 2: Forest Fire Interventions and Ecosystem Protection

During dry summer months, needle shedding in Chir Pine (*Pinus roxburghii*) forests creates flammable fuel beds. The plan integrates:

- Community pine-needle collection initiatives and controlled fuel-line burning prior to peak summer.
- Rapid-response firefighting deployments by the State Forest Department.
- Provision of artificial water troughs in wildlife sanctuaries to prevent wild animal dehydration and reduce human-wildlife conflicts.

2.3 Pillar 3: Springhead Protection and Tourist Hydration Logistics

- **Springhead Rejuvenation (*Dhara Vikas*):** Monitoring traditional natural springs (*Naulis*/*Khatris*) and deploying Jal Shakti Vibhag water tankers to high-altitude settlements facing discharge drops.

- **Floating Population Management:** Millions of seasonal tourists travel to hill stations during peak summer. The HAP integrates tourist highway advisories and hydration kiosks along key corridors (NH-05, NH-21) leading to Shimla, Manali, and Dharamshala.

2.4 Pillar 4: Agro-Horticultural Adaptation Directives

To protect apple orchards and cash crops from thermal stress:

- The Department of Horticulture issues advisories on anti-hail/anti-shade net deployment, drip micro-irrigation, and organic mulching to conserve soil moisture.

- Guidance on managing the upward elevation shift of traditional apple cultivation lines caused by loss of winter chilling hours and summer heat.

3 Seasonal Implementation Workflow

The operational workflow follows a three-phase seasonal calendar:

Table 1: Seasonal Implementation Calendar of the Himachal Pradesh HAP

Phase	Timeline	Primary Operational Tasks
Phase I: Pre-Heat	Jan – Mar	Inter-departmental coordination (HPSDMA, Forest, Health, Jal Shakti); forest fire line clearance; inventory audits for ORS/IV supplies; spring discharge baseline mapping.
Phase II: Active Heat	Apr – Jun	Real-time IMD hill forecast tracking; issuing color-coded alerts; deploying forest fire rapid response teams; managing tourist highway hydration hubs; emergency water tanking.
Phase III: Post-Heat	Jul – Sep	Wildfire damage assessments; evaluating health and hospital admission data; auditing water tanker distribution; updating agricultural risk maps.

4 Critical Challenges and Policy Outlook

- **Unplanned Urbanization & Thermal Traps:** Concrete construction and corrugated tin roofing in high-density hill towns (e.g., Shimla, Solan) create localized micro-climate urban heat traps due to the loss of traditional timber-stone passive thermal architecture.
- **Compounding Disaster Cascades:** Extreme summer dry spells and heatwaves exacerbate topsoil degradation, increasing vulnerability to flash floods and landslides during the subsequent monsoon season.
- **Water Stress During Peak Tourist Season:** The convergence of peak summer heatwaves and tourist influx places heavy demand on municipal water supply networks in high-altitude urban centers.

5 Conclusion

The Himachal Pradesh Heat Action Plan provides an adaptive model for climate resilience in high-altitude mountain ecosystems. By connecting public health protocols with forest fire management, mountain spring preservation, and agricultural advisories, the state offers a framework for managing heat risks in ecologically fragile mountain regions.